

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Test the following series for convergence or divergence.

(a) $\sum_{n=1}^{+\infty} \frac{2^n}{n!}$

Using the Ratio Test

$$\begin{aligned} \lim_{n \rightarrow +\infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow +\infty} \left| \frac{\frac{2^{n+1}}{(n+1)!}}{\frac{2^n}{n!}} \right| \\ &= \lim_{n \rightarrow +\infty} \left| \frac{2^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n} \right| \\ &= \lim_{n \rightarrow +\infty} \left| \frac{2}{n+1} \right| = 0 \end{aligned}$$

$\Rightarrow$ (absolutely) Convergent

(b) $\sum_{n=2}^{+\infty} \frac{1}{n\sqrt{\ln n}}$

$f(x) = \frac{1}{x\sqrt{\ln x}}$, decreases in $[2, +\infty)$

Use Integral Test.

$$\int_2^{+\infty} \frac{1}{x\sqrt{\ln x}} dx = \lim_{t \rightarrow +\infty} [2\sqrt{\ln x}]_{x=2}^{x=t}$$

$= +\infty$

$\Rightarrow$ divergent.

Problem 2. (6 points) Find the radius of convergence and the interval of convergence of

the power series $\sum_{n=0}^{+\infty} \frac{nx^n}{2^n}$.

$$\begin{aligned} a_n &= \frac{nx^n}{2^n} \\ \lim_{n \rightarrow +\infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow +\infty} \left| \frac{\frac{(n+1)x^{n+1}}{2^{n+1}}}{\frac{nx^n}{2^n}} \right| = \lim_{n \rightarrow +\infty} \left| \frac{(n+1)x^{n+1}}{2^{n+1}} \cdot \frac{2^n}{nx^n} \right| \\ &= \lim_{n \rightarrow +\infty} \left| \frac{(n+1)x}{2n} \right| = \left| \frac{x}{2} \right| \end{aligned}$$

By the Ratio Test, we need $\left| \frac{x}{2} \right| < 1 \Rightarrow |x| < 2 \Rightarrow R=2$

When $x=2$, $\sum_{n=0}^{+\infty} \frac{nx^n}{2^n} = \sum_{n=0}^{+\infty} n$ diverges

$x=-2$, $\sum_{n=0}^{+\infty} \frac{nx^n}{2^n} = \sum_{n=0}^{+\infty} (-1)^n n$ diverges.

Thus its interval of convergence is $(-2, 2)$